Investigating Accessibility to train stations in Greater Kuala Lumpur and providing data-driven interventions

What type of GenAI: ChatGPT, Claude

I use GenAI for assistance for:

- Verify novelty of question
- Dataset Discovery
- Literature Exploration Support
- Summarise key points of research papers
- Interpreting research findings
- Finding business models and stakeholders
- Brainstorming analytic levels and methodology
- Discovering potential stakeholder and benefits of project
- R coding syntax
- Interpreting data analysis results
- Summarising findings

Key changes based on marker feedback:

Altered *Introduction* paragraph slightly to provide more direct quantitative evidence on the measurable impact of first/last-mile barriers on public-transport as recommended by feedback.

Added a comparison table to *related work* section that highlights related studies better, as recommended by lecturer.

Updated *business model* to present more clearly the ownership and delivery of project as well as adding a paragraph for implementation challenges according to feedback.

Introduction

The first and last mile (FLM) problem is a crucial factor in underutilization of public transportation, it is the difficulty in accessing public transit stations (Lu, et al., 2024). FLM problem increases the likelihood of private car usage to access public transit stations which contributes to traffic congestion, air pollution and increases expenditure on construction and maintenance of parking facilities around stations (Lu et.al, 2024). The systematic review stated that type of public transport available and the spatial accessibility of public-transport stops are some of the factors that affect FLM mode choice.

MRT and LRT services in Greater Kuala Lumpur (GKL) provide efficient and reliable services to the population. However, despite this, Malaysia's urban transport challenges are longstanding and expanding rail infrastructure alone would be insufficient if travellers did not shift away from private vehicles (Almselati et al., 2011; Chuen et al., 2014). Public transport modal share in GKL remains at approximately 20-25%, against an official target of 70% (Shukri & Ho, 2025), and this shortfall is

disproportionately concentrated in sub-urban areas facing longer FLM connections (Shukri & Ho, 2025). A qualitative study of 34 riders found commuters describing dependence on private vehicles or ride-hailing where connections are weak (Shukri & Ho, 2025). Chin et al. (2020) documented network walking distances of 2,300-3,400m to stations in GKL that were only 398-543m away by straight line, causing walking routes to be 4 to 8.5 times longer than necessary. Yusoff et al. (2021) documented that 46% of Putrajaya residents use e-hailing and DRT services, likely substituting for inadequate first-mile connections. The Malaysian Government aims to increase public transport usage to approximately 40% by 2030 (Halim, et al., 2025). This motivates investigation into improvements of pedestrian connections, feeder buses and demand-responsive transport (DRT).

This project aims to develop a spatial decision-support framework for identifying and prioritising FLM interventions around MRT stations in GKL. This framework is created by combining (all public data), population estimates from WorldPop, pedestrian-network measurement derived from OpenStreetMap and using OSMnx python library for geospatial calculations, MRT service data, existing DRT coverage and station ridership from Official Malaysia Open Data Portal (data.gov.my) GTFS. Using this data-driven framework to identify areas with indirect walking routes, excessive walking distances or insufficient connecting services. These accessibility profiles will support provisional recommendations for pedestrian infrastructure, feeder bus service or DRT service improvements.

Related Work

Table 1

Study	Scale	Method	Findings
Chin et al. (2020)	3 LRT stations, Klang Valley	Krambeck's Global Walkability Index	A higher walkability index increases station usage. Poor connectivity and accessibility are reasons for reduced walking. Future work should add ground surveys and GIS mapping.
Ruslan et al. (2023)	Selected KL city-centre stations	Spatial Walkability Index (SWI)	SWI varied spatially across KL city centre; index was driven by expert-assigned criteria weights, but not validated against any usage or ridership outcome.
Khoo (2025)	All of GKL	Road permeability + POI accessibility	Walkability in GKL is uneven. Uses the same routing tools (OSMnx) to measure walkability.
Sunoto & Ponrahono (2019)	4 feeder routes, Kajang MRT	GIS 400m buffer, land-use composition	Kajang's feeder routes are not dense or concentrated enough. Future work should implement triangulation method for feeder route optimisation.
Halim et al. (2025)	2 feeder routes, Asia Jaya LRT	Smart-card ridership + spatial autocorrelation	Knowing when and where demand peaks can improve bus schedules, route design and resource allocation for FLM to stations.

Rahaman et al. (2025)	Puncak Alam	GIS Transit Accessibility Index (FAHP)	Authors recommend using the FAHP model to plan feeder and DRT services, with future work adding more experts and real-time GTFS data.

Table 1 summarise studies related to this project. Walkability studies (Chin et al., 2020; Ruslan et al., 2023; Khoo, 2025) measure pedestrian-network conditions but stop at the station boundary, without linking to feeder or DRT coverage. Conversely, feeder-bus studies (Sunoto & Ponrahono, 2019; Halim et al., 2025) examine service catchment and ridership determinants but do not measure how difficult the walk to the station or bus stop actually is. Rahaman et al. (2025) comes closest to a combined decision-support tool but only looked at suburban area without direct rail access. No existing GKL study measures network circuity, service coverage and population together, across the full set of MRT stations, in a form that maps directly to intervention categories.

Business Model

This project represents information-based brokering because it integrates multiple spatial and transport datasets to produce value-added accessibility profiles and decision-support information for transport planners. It helps decision-makers identify areas for pedestrian infrastructure, feeder-bus or DRT interventions. The project would be a business-to-government or business-to-business analytical service delivered by a research or consultancy provider commissioned by Prasarana, Rapid Bus or local authorities.

The project implements descriptive analysis to summarise existing FLM conditions and apply clustering algorithms using unsupervised machine learning to group residential areas into accessibility profiles. The prescriptive component will use transparent decision rules to match each accessibility profile with a suitable intervention which will be delivered as an interactive dashboard or periodic report.

This project can benefit transport operators by identifying where existing feeder routes or DRT may have insufficient coverage or frequency. For infrastructure planners, the project identifies communities that require improvements in pedestrian infrastructure, connecting paths or additional station entrances. Financially, the project can help decision-makers allocate limited budgets more efficiently. This reduces the risk of implementing expensive services in unsuitable locations.

Ownership and responsibility for implementation are, split across stakeholders: Rapid Bus manages feeder and on-demand operations, while local authorities are generally responsible for surrounding pedestrian infrastructure, so no single stakeholder can act on all recommendations independently. Data ownership also sits with the platform provider rather than the government, since the analytical layer is built on open data rather than proprietary datasets.

Implementation faces several practical constraints. OSM pedestrian-network completeness varies across GKL, and gaps or inaccuracies in footpath mapping could bias circuity estimates in less-surveyed areas. WorldPop population data is

modelled rather than directly observed, introducing uncertainty at fine spatial resolution. The framework also does not capture safety, comfort or weather protection. Hence, these conditions need to be validated against interventions.

Characterising and Analysing Data

Source	Volume	Velocity	Variety	Veracity
GTFS static	Small, a few MB per feed	Low, updated periodically by Prasarana, not real-time	Structured, txt file	High, official operator-published schedules
Rapid Rail ridership	Very large, exceeds Excel's row limit; contains daily ridership	High, updated daily	Structured, (CSV/Parquet)	High, derived from actual tap-in/tap-out transaction data
OSM pedestrian network	Large at country scale (~232MB to download), small once clipped to a station catchment	Low, a static snapshot that can be updated by contributors	Semi-structured geometry + inconsistent tagging	Veracity is uneven, OSM-mapped network varies with gaps or inaccuracies in footpath mapping. Data is crowd sourced.
WorldPop population	Moderate ~ 100 million cells	Low, Updated annually/periodically, not real-time	Structured, gridded numeric	Moderate, estimated population, not directly observed

R is used since it natively supports geospatial vector data (roads, boundaries), gridded raster data (population), and tabular transit data (GTFS) within one environment. Local processing on a standard laptop (32GB RAM) is sufficient because data volume is deliberately scoped down at extraction.

Building and cleaning the pedestrian network, and computing shortest-path circuity for each residential area, are the most computationally intensive steps, since each station required an independent routing operation rather than a simple lookup. Extending this to all MRT/LRT/feeder stations across GKL would require changes to data processing and storage strategies. For processing speed, parallelized computation (e.g. R's furrr package) would let independent per-station calculations run concurrently rather than sequentially, addressing the roughly linear runtime scaling. For storage, moving from flat files to a spatial database such as PostGIS would allow station-level results to be queried and updated incrementally as new GTFS or OSM data becomes available, rather than reprocessing the full dataset each time. For delivery, a live tool such as R Shiny would let stakeholders query results interactively, supporting the dashboard-based delivery model, rather than requiring a new static report for each update.

Data Analysis Approach

Catchment radii of 800m for MRT stations and 400m for feeder/rapid bus stops follow PLANMalaysia's Green Neighbourhood Guidelines and established

international conventions (El-Geneidy et al., 2014; Guerra et al., 2012). However, Sukor and Fisal (2018) found Penang bus users' average walking distance to be 423m, while Azmi and Karim (2012) found residents in Shah Alam walked 200m to community facilities, but residents in single-use zoned areas, were largely unwilling to walk regardless of distance. This suggests catchment behaviour is shaped as much by density and land-use mix as by distance alone, and these standard radii should be understood as planning conventions rather than fixed empirical thresholds.

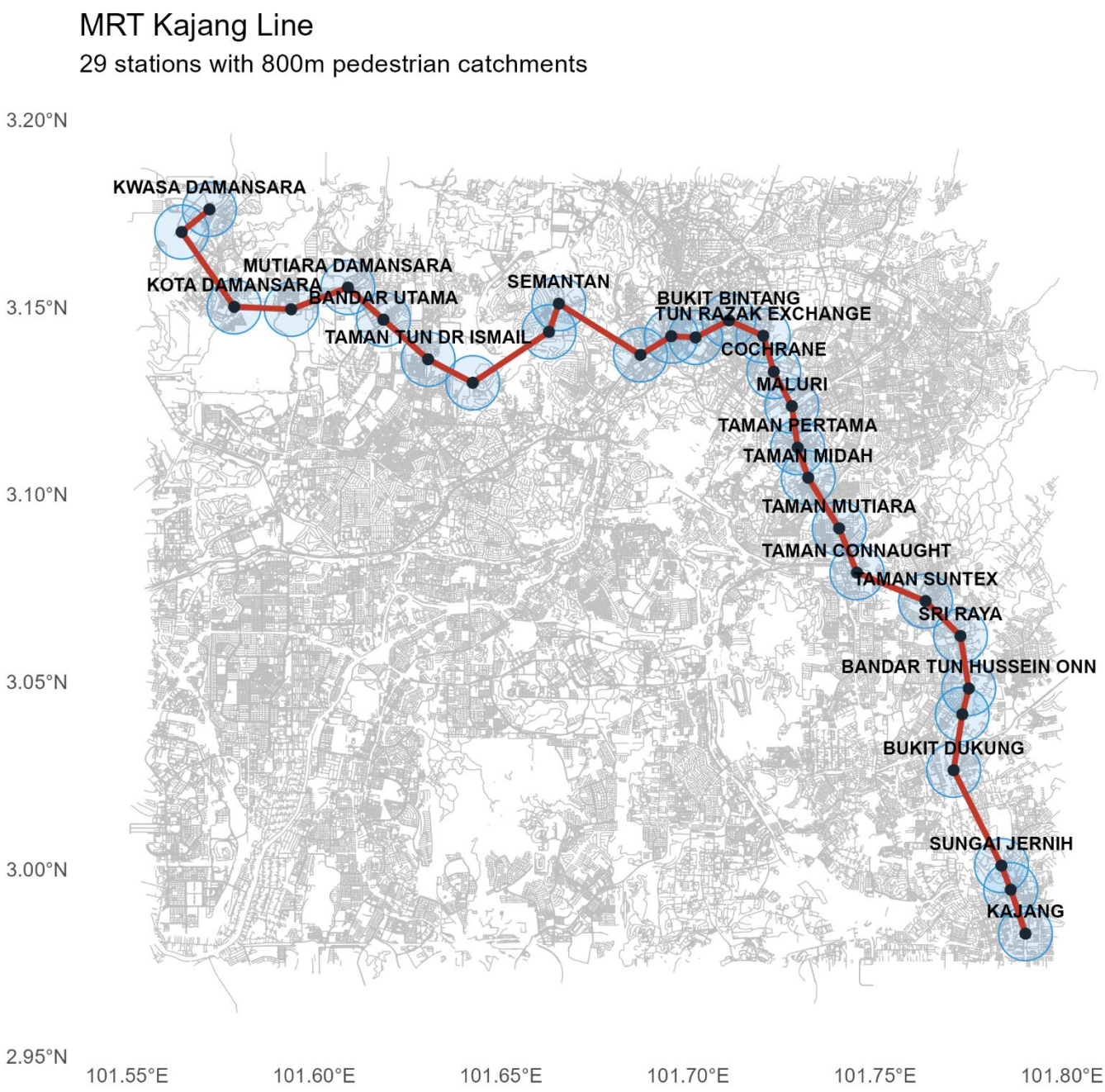
Station names and coordinates are extracted from GTFS static feeds. Using OpenStreetMap data with R's sf, osmextract and sfnetworks packages, residential areas within 800m of each MRT station are identified, and the network and straight-line distances from each station to each residential centroid are calculated, their ratio gives circuity, with network distance routed through OSM pedestrian paths. The same routable network is reused to measure bus accessibility, residential areas within 400m of a bus/feeder stop are identified and circuity calculated. Centroids with no stop within 400m are recorded as having no feasible feeder alternative rather than poor circuity. DRT coverage is treated qualitatively, as no structured public dataset exists for Malaysia's DRT services. Population is directly extracted from the WorldPop raster over the 800m catchment area of each station.

Station-level ridership is extracted from data.gov.my's origin-destination dataset, which records trip counts since 2023. A correlation test between circuity and average daily ridership tests whether poorer pedestrian access is associated with reduced station usage. A simple regression of ridership on population produces an expected ridership figure per station, and the gap between actual and expected ridership helps indicate how much circuity itself affects usage. A station with high population, low ridership relative to that expectation, and poor circuity suggests access is genuinely suppressing demand and warrants intervention.

Each station, bus and feeder’s mean circuity, population, and ridership gap are aggregated to station level and used as inputs to unsupervised clustering, grouping stations into accessibility profiles. The number of clusters is selected using the elbow method or silhouette score rather than an arbitrary choice. Each resulting cluster is then matched to a candidate intervention (pedestrian infrastructure improvements, bus route extension, or DRT deployment).

Demonstration

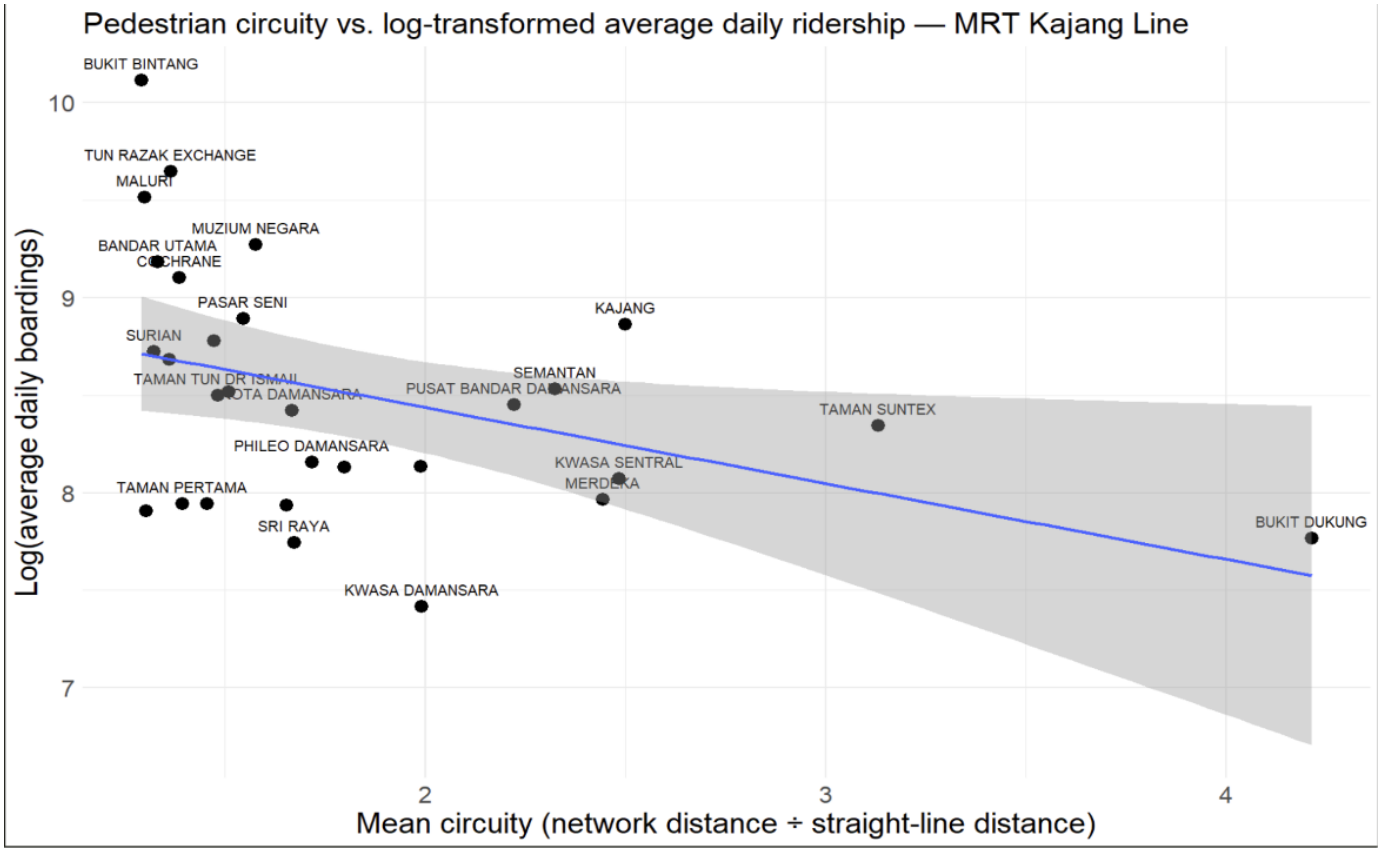
This demonstration applies part of the analysis approach described above. Only the MRT Kajang Line will be investigated and the data analysed are pedestrian circuity, population, ridership validation, and clustering. This demonstration excludes the feeder and rapid bus stops circuity analysis and only uses station ridership data from 2026. Processing required resolving topological gaps in the OSM pedestrian network via coordinate snapping, a known limitation of crowd-sourced map data. All 29 stations were successfully processed within approximately 1 hour on a standard laptop.



Checking for semantic validation, only one notable error was found. Five residential areas returned circuity below 1, a value below the theoretical minimum, arising when a residential area sits very close to its station and node-snapping introduces distance smaller than the true straight-line separation. These cases were retained but noted as a limitation of the snapping approximation at short distances.

Mean circuity across the 29 stations ranged from 1.29 to 4.21 (mean = 1.82). A correlation test between circuity and log-transformed average daily ridership found a statistically significant negative relationship ($r = -0.40$, $p = 0.031$), consistent with the premise that poorer pedestrian access is associated with reduced station usage.

Figure 1 visualises this correlation with a scatter plot of log average daily ridership against average circuity.



A regression of ridership on population explained very little variance ($R^2 = 0.021$) and the population coefficient was not statistically significant ($p = 0.453$), indicating that population alone is a poor predictor of station-level-ridership. This is consistent with Guerra et al. (2012), who found that ridership is more strongly driven by nearby employment than by nearby population. Interchange stations, whose high ridership is influenced by factors such as tourism and employment density unrelated to pedestrian access could be the reason why population poorly predicts ridership and why the circuity-ridership correlation weakens when these stations are excluded ($r = -0.40$ vs. $r = -0.30$ excluding interchanges).

Stations were grouped using ward's hierarchical clustering on mean circuity, catchment population and the log ridership gap (ridership relative to the population-based expectation). Silhouette widths were weak for every k from 2 to 8 (0.25–0.28), indicating no sharply separated groups, so silhouette was used as a guide. K=5 was selected because the clusters were mapped to distinct intervention types: low-circuity, high-ridership indicating low-priority for intervention. Low-circuity, low-ridership prompting investigations for low station usage despite low-circuity. High-circuity, high-population and high-circuity, low-population indicating high priority for the prior and finally extremely high-circuity (4.21) with low-ridership which was one station, indicating urgent investigation for such an outlier.

Clustering results showed some sensitivity to station selection and random initialization, consistent with the limited sample size of a single line. Extending the analysis to the Putrajaya LRT and MRT lines, and incorporating the feeder/rapid bus circuity described earlier but not built in this demonstration, would both increase sample size, likely stabilising the clustering results, and complete the proposed feature set.

Several directions for future investigation emerged from this process. Adding employment density, as a factor, may better predict ridership. Interchange stations warrant separate treatment, since their ridership appears to be driven by network-wide attraction rather than local access. Additionally, modelled rather than observed population data and crowd-sourced pedestrian network data of uneven completeness was used. Meaning decisions informed by this analysis should be validated against local knowledge or site inspection before any intervention recommendation is acted upon. This demonstration shows the proposed method is feasible on real data. Using the proposed additional data and performing robust EDA would greatly improve and refine this study.

Data Link: Google Drive

Standards, Data Governance and Management

This project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, structuring work through business understanding (defining the first/last-mile accessibility problem), data understanding and preparation (the 4 V's characterisation and extraction pipeline described above), modelling (circuity calculation and clustering), evaluation (validation against ridership data) and deployment (dashboard-based delivery model).

All datasets used are publicly accessible under open licences: GTFS static feeds and ridership data are sourced from data.gov.my under Malaysia's Public Sector Open Data licence; OpenStreetMap data is licensed under the Open Database Licence; WorldPop population data is licensed under Creative Commons Attribution 4.0. No proprietary or restricted-access data sources are used, meaning the project's outputs could in principle be reproduced or extended by any interested stakeholder without needing special data-access agreements.

No personally identifiable information is processed. Ridership data does not identify individual travellers and population data is a modelled gridded estimate. Since the data represent estimates rather than directly observed populations and areas with incomplete OSM mapping risk being systematically under- or over-represented in the results, regardless of their actual on-the-ground conditions. These uncertainties should be acknowledged when interpreting the results.

References

Almselati, A. S. I., Rahmat, R. A. O. K., & Jaafar, O. (2011). An overview of urban transport in Malaysia. *Social Sci, 6*(1), 24–33.

Azmi, D. I., & Karim, H. A. (2012). A comparative study of walking behavior to community facilities in low-cost and medium cost housing. *Procedia – Social and Behavioral Sciences, 35*, 619–628. https://doi.org/10.1016/j.sbspro.2012.02.129

Chin, S. J. Y., Goh, B. H., Teo, F. Y., Kang, B. G., & Yuen, C. W. (2020). Pedestrian walkability index for light rail transit (LRT) stations in Klang Valley, Malaysia. In F. M. Nazri (Ed.), *Proceedings of AICCE'19: Transforming the nation for a sustainable tomorrow* (pp. 737–748). Springer. https://doi.org/10.1007/978-3-030-32816-0_52

Chiu Chuen, O., Karim, M. R., & Yusoff, S. (2014). Mode choice between private and public transport in Klang Valley, Malaysia. *The Scientific World Journal, 2014*(1), 394587. https://doi.org/10.1155/2014/394587

Department of Statistics Malaysia. (2020). *Malaysia population count, 100m resolution* [Data set]. WorldPop. University of Southampton. https://hub.worldpop.org/geodata/listing?id=29

El-Geneidy, A., Grimsrud, M., Wasfi, R., Tétreault, P., & Surprenant-Legault, J. (2014). New evidence on walking distances to transit stops: Identifying redundancies and gaps using variable service areas. *Transportation, 41*(1), 193–210. https://doi.org/10.1007/s11116-013-9508-z

Geofabrik GmbH. (2026). *OpenStreetMap data extracts: Malaysia, Singapore, Brunei* [Data set]. https://download.geofabrik.de/asia/malaysia-singapore-brunei.html

Gregory Ho, W. S., Shukri Mohamed Khairi, & Ling, K. S. S. (2025). *Closing the first-and-last mile gap in Greater Kuala Lumpur*. Khazanah Research Institute.

Guerra, E., Cervero, R., & Tischler, D. (2012). Half-mile circle: Does it best represent transit station catchments? *Transportation Research Record, 2276*(1), 101–109. https://doi.org/10.3141/2276-12

Halim, M. A., Rosni, N. A., & Tsong, T. B. (2025). Preliminary insights into feeder bus usage: A case study of Asia Jaya LRT station in the Klang Valley. *Journal of Design and Built Environment*, 74–88.

Khoo, W. Y. (2025). *Measuring walkability of Greater Kuala Lumpur*. Khazanah Research Institute.

Lu, Y., Kimpton, A., Prato, C. G., Sipe, N., & Corcoran, J. (2024). First and last mile travel mode choice: A systematic review of the empirical literature. *International Journal of Sustainable Transportation*, *18*(1), 1–14. https://doi.org/10.1080/15568318.2023.2218285

Ministry of Transport Malaysia. (2026). *Rapid Rail ridership (origin-destination), daily* [Data set]. data.gov.my. https://data.gov.my/data-catalogue/ridership_od_rapidrail_daily

Prasarana Malaysia Berhad. (2026). *GTFS static feed: Rapid Rail KL* [Data set]. data.gov.my. https://api.data.gov.my/gtfs-static/prasarana?category=rapid-rail-kl

Rahaman, L. A., Naharudin, N., Salleh, S. A., & Alias, E. (2025). A multi-criteria GIS approach to evaluate transit accessibility index: Integrating FAHP in suburban Malaysia. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 48*, 255–263. https://doi.org/10.5194/isprs-archives-XLVIII-4-W6-2025-255-2025

Ruslan, N., Naharudin, N., Salleh, A. H., Halim, M. A., & Abd Latif, Z. (2023). Spatial walkability index (SWI) of pedestrian access to rail transit station in Kuala Lumpur city center. *Planning Malaysia, 21*.

Shukri Mohamed Khairi, & Gregory Ho, W. S. (2025). *Greater Kuala Lumpur's public transportation and its viability: A qualitative study*. Khazanah Research Institute.

Sukor, N. S. A., & Fisal, S. F. M. (2018). Factors influencing the willingness to walk to the bus stops in Penang Island. *Planning Malaysia, 16*(1), 193–204. https://doi.org/10.21837/pm.v16i5.423

Sunoto, Y. N., & Ponrahono*, Z. (2019). Assessment Of Mrt Feeder Bus Routes' Catchment Area Using Geographical Information System. In P. A. J. Wahid, P. I. D. A. Aziz Abdul Samad, P. D. S. Sheikh Ahmad, & A. P. D. P. Pujinda (Eds.), Carving The Future Built Environment: Environmental, Economic And Social Resilience, vol 2. European Proceedings of Multidisciplinary Sciences (pp. 481-489). Future Academy. https://doi.org/10.15405/epms.2019.12.47

Yusoff, I., Ng, B.-K., & Azizan, S. A. (2021). Towards sustainable transport policy framework: A rail-based transit system in Klang Valley, Malaysia. *PLOS ONE, 16*(3), e0248519. https://doi.org/10.1371/journal.pone.0248519